

A Preliminary Loss Comparison of Solid-State Transformers in a Rail Application Employing Silicon Carbide (SiC) MOSFET Switches

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Abstract

The silicon carbide (SiC) power semiconductor and the medium frequency transformer (MFT) are two enabling technologies that are allowing the efficiencies and power densities of power electronic converters to be pushed beyond the limits of those that have previously been possible; the solid-state transformer (SST), with its viability based in part upon the two aforementioned technologies, has much potential to offer in Rail, Marine, Aviation and Smart-Grid applications due to its high performance operation and compact form-factor and so it is studied in-depth in this paper, where a comparison of the active and passive component losses between various SST topologies for a rail traction application, modelled with 3.3 kV and 10 kV SiC MOSFET switches alongside benchmark 3.3 kV and 6.5 kV Si IGBT switches, operating on a 50 Hz, 25 kV ac, overhead supply has been made. In addition, a comparative analysis is made of the estimated costs of the converters studied.

1 Introduction

The modular structure of the SST enables power switches to be used with forward voltage ratings which were not before possible. The advancement in SiC-based power switches, in particular in the commercially realised MOSFET, has led to an increase in the break down voltage ratings of devices, such as the commercially ready 10 kV and 3.3 kV SiC die [1]. The combination of these two facts, amongst others, has led to an increase in the viability of the multilevel SST [2]. In addition to this, the switching losses in the semiconductor devices can be reduced due to the properties of the SiC material [3]. The commercial readiness of the SiC power MOSFET and the SiC JBS diode is the reason for why they are prioritised in this study above other SiC device constructions, such as the SiC IGBT or BJT.

For rail traction drives, the topology traditionally used is a single-phase AC-DC converter galvanically isolated from the rail supply by a line frequency multi-tapped step-down transformer. Though offering good levels of efficiency and reliability, the converter suffers from a bulky transformer due

to operation at line frequency, which leads to a low converter power to weight ratio. Introducing an MFT-based DC-DC converter into the topology has the potential to allow the size of the magnetics to be greatly reduced. The highly efficient range of converters based upon the dual active bridge, in combination with the weight savings in the transformer, make a strong case for considering the SST as alternative to the existing traction drive technology [2]. Modularisation through input series and output parallel (ISOP) connection of the H-bridge cells is another common and attractive feature in SSTs, enabling voltage sharing at the input and current sharing at the output, and so the converters that are studied in this paper combine a cascaded H-bridge active front end converter (AFEC) with a medium frequency DC-DC output stage, shown in Figure 1.

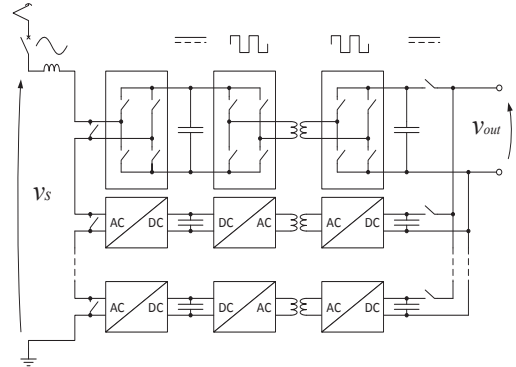


Figure 1: An n -level SST with a cascaded H-bridge AC-DC input stage and an MFT-based DC-DC output stage

To make a detailed comparison of the losses in the four different converter topologies the passive component losses have been calculated as well as the switching and conduction losses of the active devices. The process for calculating the losses included building semiconductor models from die data, basic dimensioning of the line inductor, the intermediate DC-link capacitors, the MFT and the DC-link capacitor, etc., and is described in Sections 3 and 4. This design work also allowed a more closely representative figure for the respective component costs to be calculated. So, a comparison is made here, with respect to the losses and relative component cost between the four different topologies using 10 kV and 3.3 kV rated SiC switches. In addition, the financial implications of using an anti-parallel diode in the SiC configurations has been explored.

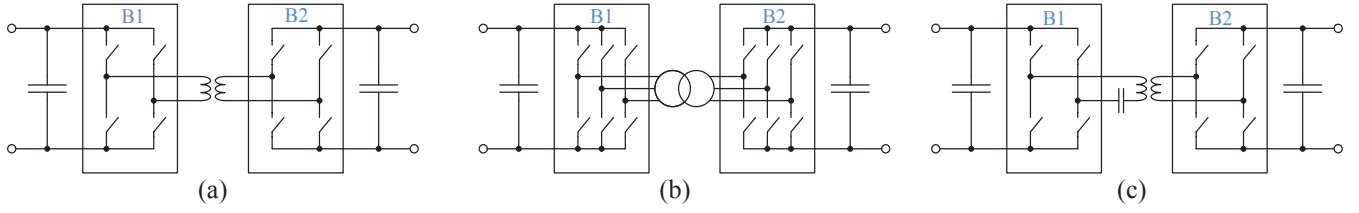


Figure 2 SST DC-DC converter configuration schemes: (a) is the DAB; (b) is the DAB3; and, (c) is the RSAB/ RDAB

2 SST Topologies for Rail

The targeted application in Rail for the SST studied here is the Electrical Multiple Unit (EMU), a class of train where the power electronics for the traction drive are distributed throughout the train instead of being located mainly in an engine car. The benefits of the said distribution of power electronics is that more room can be made on board for passengers, due to the modularity and potential increase in power density per volume of the SST, and that the rolling weight of the train may be reduced, due to the increase in power density by weight.

The AC-DC input stage of the proposed SST is based upon a CHB architecture and is assumed to be operated using phase-shifted pulse-width modulation. The topology was chosen for its ability to distribute the supply voltage equally over the switching devices connected in series, the reduction that it can offer in the supply current and multi-level voltage waveforms and also because of the redundancy which can be incorporated into the modular multi-level structure. The PWM strategy has been chosen because of its balanced distribution of switching and conduction losses of the semiconductors in the topology [4].

There are four different DC-DC topologies of interest in this losses study: the non-resonant dual-active bridge (DAB); the three-phase non-resonant dual-active bridge (DAB3); the series-resonant dual-active bridge (RDAB); and, the series-resonant single-active bridge (RSAB). This paper makes a comparison between the AFEC in combination with each of the four aforementioned DC-DC converter topologies. All DC-DC converter topologies in the study were modulated by phase-shifted square wave modulation, except for the SRAB, which was modelled using voltage modulation.

3 Converter Dimensioning

The fixed parameters used in the simulation of the configurations of all of SSTs studied are listed in Table 1.

Table 1 - The SST fixed input parameters: supply voltage peak, line frequency, input power, absorbed reactive power, and the output DC voltage.

Parameter	$\hat{v}_{s,max}$	f_{line}	P_{in}	Q_{in}	V_{out}
Value	28 kV	50 Hz	2 MW	10 kVAr	1.8 kV

3.1 SST Configurations

There were four configurations of SST chosen for this study, categorised here according to the semiconductors selected. The nominal value of V_{out} determined that a 3.3 kV rated semiconductor should be used for DC-DC bridge 2 in all of the converter models.

Table 2 - SST simulation configurations and device ratings.

Config	Device	AFEC	DC-DC B1	DC-DC B2
1	SiC MOS	3.3 kV	3.3 kV	3.3 kV
2	Si IGBT	3.3 kV	3.3 kV	3.3 kV
3	SiC MOS	10 kV	10 kV	3.3 kV
4	Si IGBT	6.5 kV	6.5 kV	3.3 kV

The semiconductor models have been derated in the simulations with respect to the device blocking voltage and also to the conduction current. From $\hat{v}_{s,max}$, V_{out} and the derated semiconductor voltage value we can fix the number of physical levels, $n_{levels,physical}$, required in the SST using (1).

$$n_{levels,physical} = \hat{v}_{s,max}/V_{nom,device} \quad (1)$$

Table 3 - SST configuration, device voltage derating and number of converter physical levels.

Config	$V_{nom,device}$			$n_{levels,physical}$
	AFEC	DC-DC B1	DC-DC B2	
1	1.8 kV	1.8 kV	1.8 kV	17
2	1.8 kV	1.8 kV	1.8 kV	17
3	6 kV	6 kV	1.8 kV	5
4	4.3 kV	4.3 kV	1.8 kV	7

For example, Configuration 1 RDAB was a 17-level SST with an RDAB DC-DC conversion stage modelled with 3.3 kV SiC MOSFET switches used throughout.

3.2 Active Front End Converter

The switching frequency of the AFEC was selected according to the guidelines in [5] and was chosen to be 450 Hz. The AFEC analytical model was derived from the phasor model of the AFEC. The resistance of the line inductor was neglected in the analysis. Using trigonometry to analyse the phasor relationship between the supply voltage, the supply current, the reactance, the multilevel voltage, and the power factor angle, the RMS multilevel voltage can be found by observing Figure 3.

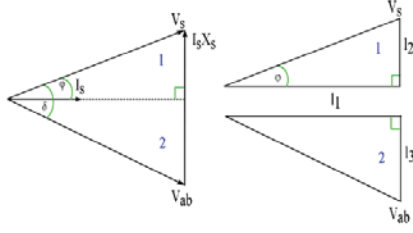


Figure 3 - Phasoral relationship between supply voltage, supply current, supply reactance, and the power factor.

$$l_3 = I_{s(rms)} \cdot X_s - V_{s(rms)} \cdot \sin \varphi \text{ [V]} \quad (1)$$

$$V_{AB(rms)} = \sqrt{(l_3^2 + (V_{s(rms)} \cdot \cos \varphi)^2)} \text{ [V]} \quad (2)$$

$V_{AB(rms)}$ is the RMS value of the voltage seen between the input terminal of the first in the series of H-bridges in the AFEC and the output terminal of the last.

A. Semiconductors – The model for the devices were built by mapping the points on the die Eloss vs I_c and Eloss vs V_{dc} curves, after which a second order polynomial curve was fitted to the respective data sets. The number of die required was estimated by the following equation, where the effective surface area of the die has been neglected:

$$n_{parallel\ die} = \frac{I_c \cdot k_{derate}}{\hat{I}_{chip,nom}} \quad (3)$$

Where k_{derate} is a current derating factor used to provide a margin of safety which will keep the device working within a safe operating area. Here a factor of 1.5 has been used. The semiconductors were also derated according to the blocking voltage. The die numbers used in the switch models are given in Table 4.

The equivalent on-state resistance, $R_{ds,device}$, of the semiconductor model was found by (4).

$$R_{ds,device} = \frac{R_{ds,die}}{n_{parallel\ die}} \text{ [\Omega]} \quad (4)$$

B. Line Inductance – The value of the inductance was set to be less than a specified value of ripple value seen in the inductor current [6].

$$L_{line} \geq \frac{1}{RF_{sw}} \sqrt{\frac{\pi}{3} \left[\frac{\pi}{4} \left(1 + \frac{3}{4} m_a^2 \right) - \frac{3}{4} m_a \right] \cdot \frac{T_s}{T}} \text{ [H]} \quad (5)$$

Where L_{line} is the line inductance, RF_{sw} is the ripple factor, T_s is the switching period and T is the period of the ac supply. Because of the low frequency of the line, 50 Hz, the material chosen for the core model was silicon steel. The type of winding chosen for the simulation was foil. A value of 3.4 mH was chosen for the inductance.

C. Intermediate DC-Link Capacitors – The construction of capacitor chosen for the intermediate DC-link capacitor was that of metal film, due to its ability to withstand high voltages as well as its purported lifespan.

3.3 DC-DC Converter

The dimensioning of the DC-DC converters with respect to the leakage inductance, the value of resonant capacitor used, and the switching frequency followed the method which is described in [7]. The switching frequency chosen was 20 kHz.

A. Semiconductors – The semiconductor models were built in the same manner as in Section 3.2 A, but instead using the rms current of the DC-DC primary and secondary side bridges in (10) rather than the rms supply current used in the AFEC model.

B. Medium Frequency Transformer – The transformer core material used in the simulation is ferrite and the windings were Litz.

4 Loss Analysis

The losses in the line inductor, AFEC switching devices, intermediate DC-link, DC-DC converter switches and the MFT were calculated using the method outlined in this section.

4.1 Semiconductor

A. Conduction Losses – equations (6) to (8) were used to calculate the conduction losses in the AFEC and DC-DC converters.

$$P_{CondMOSFET} = I_{c,(rms)}^2 \cdot R_{DS,on} \text{ [W]} \quad (6)$$

Where $R_{DS,on}$ is the device on-state resistance.

$$P_{CondIGBT} = I_{c,(average)} \cdot V_{CE0} + I_{c,(rms)}^2 \cdot R_{ce} \text{ [W]} \quad (7)$$

Where V_{CE0} is the on-state zero-current collector-emitter voltage and R_{CE} is the device on-state resistance.

$$P_{CondDiode} = I_{c,(average)} \cdot V_{F0} + I_{c,(rms)}^2 \cdot R_{D,on} \text{ [W]} \quad (8)$$

Where V_{F0} is the on-state zero-current forward voltage and $R_{D,on}$ is the device on-state resistance.

B. Switching Losses – equation (9) was used to calculate the switching losses in the AFEC and DC-DC converters.

$$P_{sw} = \frac{1}{\pi} \cdot [E_{on}(I_c) + E_{off}(I_c)] \cdot f_{sw} \text{ [W]} \quad (9)$$

Where I_c is the device current and second order polynomials were used to fit the energy loss data of the device in the transient turn-on and turn-off conditions.

4.2 Active Front Converter

The line inductor losses were calculated based upon the assumption that it was the fundamental of the current

waveform that generated the losses in the core and the windings.

A. *Line Inductance Copper Losses* – The Dowell equation [8] is widely used to calculate the increase of the alternative resistance compared to the continuous resistance (10).

$$Fr = \frac{R_{AC}}{R_{DC}} = X' \cdot \left[\frac{\sinh(2X') + \sin(2X')}{\cosh(2X') - \cos(2X')} + \frac{2}{3}(m^2 - 1) \frac{\sinh(2X') - \sin(2X')}{\cosh(2X') + \cos(2X')} \right] \quad (10)$$

Where $X' = d/\delta \cdot \sqrt{\eta_w}$ is the ratio between the conductor or foil thickness (d), the skin depth (δ), η_w is the filling factor of the winding height and m is the number of layers.

B. *Line Inductance Core Losses* – The Steinmetz equation [9] widely used to calculate the core loss for a sinusoidal magnetic induction is given by (11). It uses some constants parameters (k, a, b) which are determined by curve fitting, these parameters are given in the Table 6.

$$P_{core} = k \cdot (f/1000)^a \cdot B_m^b \cdot \rho \quad [W] \quad (11)$$

C. *Intermediate DC-Link Capacitor ESR Losses* –The rms value of the capacitor current was found by using (12).

$$I_{cap,(rms)} = \quad (12)$$

$$\hat{I}_s \sqrt{\frac{m_a}{24 \cdot \pi} [24 - 6 \cdot \pi \cdot m_a + (8 - 3 \cdot \pi \cdot m_a) \cdot \cos 2\phi]} \quad [A]$$

$$P_{cap,loss} = \frac{\tan \delta}{\omega \cdot C} \cdot I_{cap,(rms)}^2 \quad [W] \quad (13)$$

Where $\tan \delta$ is the tangent of the loss angle, ω is the angular frequency of the device current and C is the capacitance of the device.

4.3 DC-DC Converter

A. *MFT Copper Losses* – In order to calculate the increase of the alternative resistance compared to the continuous resistance for Litz wires, the equation used is given in [10].

B. *MFT Core Losses* – In this magnetic power device, the magnetic induction is triangular so the previous equation (11) can't be used. In this case, the Improved Generalized Steinmetz Equation (IGSE) [11] can be used because it takes in account the dB/dt , in order to calculate the core loss. It uses the same constants parameters (k, a, b) which are determined by curve fitting, these parameters are given in Table 5.

Table 5 - Inductor core (FeSi) and MFT (Ferrite) parameters.

Parameter	k	a	b	B _{max}
FeSi	150	1.79	2.06	1.7 T
Ferrite	1.134	1.51	2.75	0.2 T

5 Results

5.1 Active Front End Converter

A. *Semiconductor* – The semiconductor data used in the simulations is given in

Table 4. The results for the losses in the AFEC are given in Figure 4.

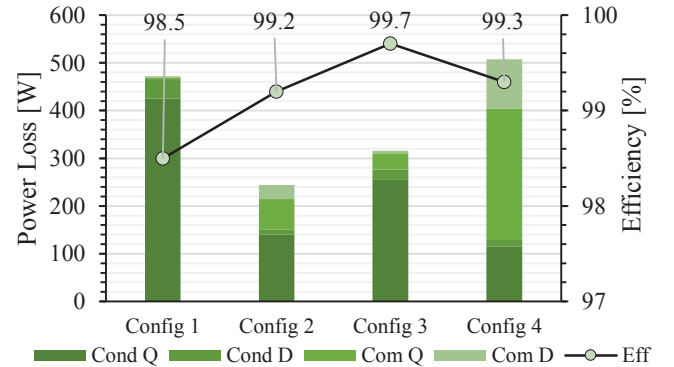


Figure 4 - AFEC semiconductor single device losses and converter efficiency.

Table 4 - Semiconductor data: device; manufacturer; voltage and current ratings; and, the number of die used in the switch models for the different converter configurations. Note: '*' denotes a SiC 3.3 kV die in Configuration 3.

Config	Device	Manufacturer/ Device type	VDS/ VCES	Ic	AFEC Die no.	DAB Die no.		DAB3 Die no.		RDAB Die no.		RSAB Die no.	
						B1	B2	B1	B2	B1	B2	B1	B2
1	CPM3-3300-00xx Z-FET	Cree/ SiC MOSFET	3.3 kV	45 A/ die	3	3	3	3	3	3	3	2	2
	CPW3-3300-Z045B	Cree/ SiC JBS diode	3.3 kV	45 A/ die	3	3	3	3	3	3	3	2	2
2	FF200R33KF2C – Half bridge	Infineon/ IGBT + diode	3.3 kV	200 A/ device									
3	CPM3-10000-0350 Z-FET	Cree/ SiC MOSFET	10 kV	15 A/ die	7	5	6*	4	4*	5	6*	5	2*
	CPW3-10000- Z015B-ES	Cree/ SiC JBS diode	10 kV	15 A/ die	7	5	6*	4	4*	5	6*	5	2*
4	FZ250R65KE3 - Single switch	Infineon/ IGBT + diode	6.5 kV	250 A/ device									

B. *Line Inductor* – A two layer core type inductor core was modelled in the simulations. The estimated efficiency of the inductor was 99.5 %.

C. *Intermediate DC-link* – The capacitor efficiencies for all configurations were 99.9 %.

5.2 DC-DC Converter

A. *Semiconductor* – The semiconductor losses in the different DC-DC converter configurations are given Figure 5 to 8.

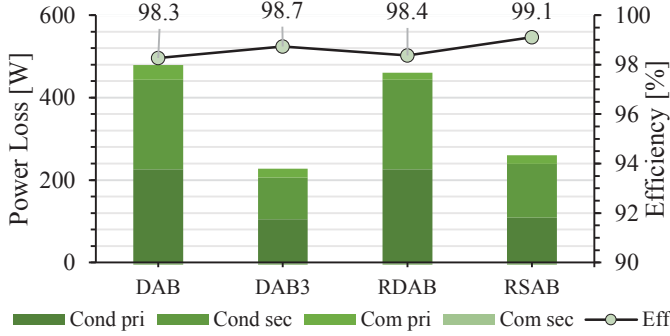


Figure 5 - DC-DC Configuration 1 (SiC, all 3.3 kV) semiconductor single device losses and converter efficiency.

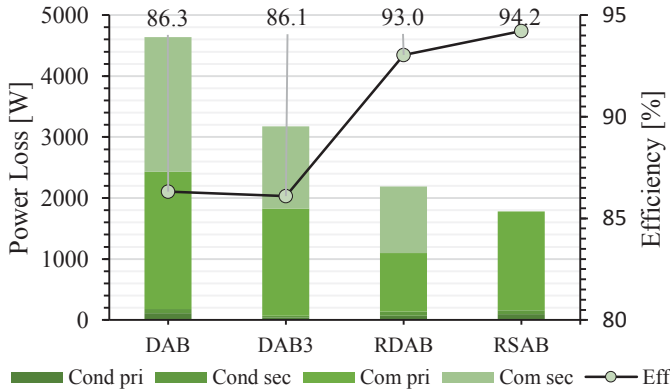


Figure 6 - DC-DC Configuration 2 (Si, all 3.3 kV) semiconductor single device losses and converter efficiency.

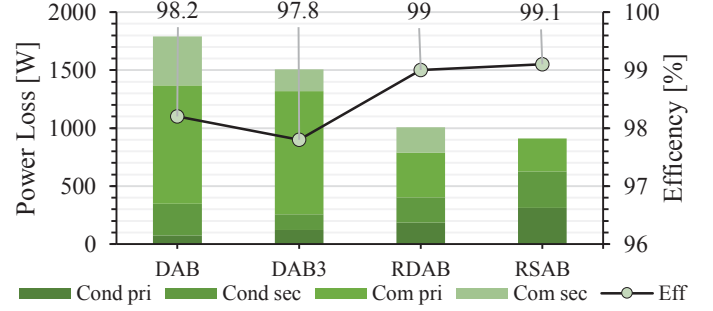


Figure 7 - DC-DC Configuration 3 (SiC, 10 kV and 3.3 kV) semiconductor single device losses and converter efficiency.

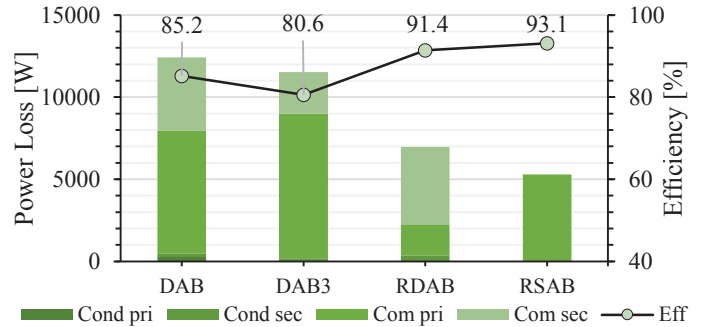


Figure 8 - DC-DC Configuration 4 (Si, 6.5 kV and 3.3 kV) semiconductor single device losses and converter efficiency.

B. *Medium Frequency Transformer* – A core type transformer model was simulated. The results are given in Table 6. The results for the DAB3 MFT were omitted from this paper due to a possible lack of validity.

5.3 SST Combined Power Electronic Losses

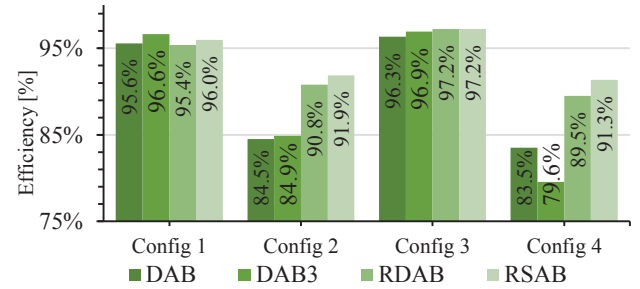


Figure 9 - SST total power electronic efficiencies; AFEC in combination with the DC-DC configurations, DC-link capacitors and the MFT.

Table 6 – Medium frequency transformer turns ratio, core volume and efficiency.

Config	1/2			3			4		
	DAB	RDAB	RSAB	DAB	RDAB	RSAB	DAB	RDAB	RSAB
$n (V_{sec} / V_{pri})$	1.02	1.02	1.02	0.30	0.30	0.27	0.42	0.42	0.41
Turns primary	10	9	11	16	13	28	27	15	27
Turns secondary	10	9	11	5	4	8	12	7	11
Volume [m ³]	0.0044	0.0048	0.0072	0.014	0.017	0.009	0.0065	0.01	0.005
η [%]	99.3	99.0	98.9	99.0	99.1	99.0	99.3	99.2	99.4

5.4 SST Relative Component Cost Estimation

A cost estimation was carried out accounting for the total cost of the semiconductors, estimated in the case of the SiC diodes only upon the die quantity used and not in fact for packaging, and also of the intermediate DC-link capacitors. For the SiC MOSFET there is a trend towards using the intrinsic body diode for freewheeling functions, instead of an additional anti-parallel diode. Although there is a possible reduction in the efficiency of a converter with MOSFET switches, due to increased switching and conduction losses, and a possible decrease in switch reliability, there are potential benefits to gain in terms of reduced material costs and decreased converter size (cf. Figure 10).

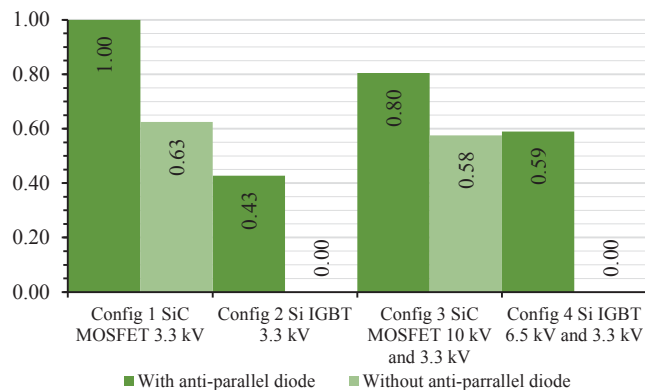


Figure 10 - Estimated relative cost of the different converter configurations with respect to the most expensive, taking into account the cost with and without an anti-parallel diode, in the case of the SiC MOSFET configurations.

6 Conclusion

The results in the study indicated that the AFEC, at the moderate switching frequency modelled, had comparable efficiencies in all of the configurations studied. For the DC-DC converters, the benefits of using SiC devices, in terms of efficiency, were apparent; at the chosen switching frequency, the efficiency of the SiC-based DC-DC converters were considerably higher, with the RSAB offering the highest overall efficiency in all cases, except for Config 1 where the omitted MFT efficiency in Config 1B skews the results. There was very little difference in the simulation data between the efficiency of the different SiC converter configurations. With respect to the estimated costs, the cost of the silicon-based solutions, in particular the 3.3 kV configurations, were considerably lower than the SiC-based configurations when the latter was using an anti-parallel diode. When no anti-parallel diode was included in the model of the SiC-based converters, the relative price difference dropped substantially. The increase in efficiency of the SiC-based converters, and the consequent decrease in operational costs may negate this price difference.

Further research on SiC MOSFET body diode reliability and also research into the benefits of using a Si IGBT-based AFEC combined with a SiC MOSFET-based DC-DC stage in the SST would help to expand this work further.

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